

Name [REDACTED]

Date March 27, 2025

Total
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- ① Let $f'(x) = \frac{(x-1)}{(x+1)}$ and $g'(x) = \frac{(x+2)}{(x-2)}$. For what values of x are they both increasing?

$$f'(x) = \frac{(x-1)}{(x+1)} \quad f'(x) = \frac{(x+2)}{(x-2)}$$

are $f'(x)$ and $g'(x)$ ↑

CP: $x = 1, -1$ CP: $x = 2, -2$

∴ they both increase $(-\infty, -1)$ and $(2, \infty)$

$f'(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
	(+)	(-)	(+)	(+)	(-)	(+)
$f(x)$	INC	DEC	INC	INC	DEC	INC

2. Use the Second Derivative Test to classify all extrema for $f(x) = 8x^3 + 81x^2 - 42x - 8$. Provide exact points.

$$f'(x) = 24x^2 + 162x - 42$$

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$$= 6(4x^2 + 27x - 7)$$

$$= 6(4x - 1)(x + 7)$$

$$x = \frac{1}{4}, -7$$

$$f''(x) = 48x + 162$$

$$f''(\frac{1}{4}) = 174 \leftarrow (+)$$

$$f''(-7) = -174 \leftarrow (-)$$

$$f(\frac{1}{4}) = -\frac{213}{16}$$

$$f(-7) = 1511$$

∴ $P(\frac{1}{4}, -\frac{213}{16})$ is a local min and $P(-7, 1511)$ is a local max

3. Given $h(x) = \frac{4x^2 - 4x + 5}{2x - 1}$, determine the equation of the oblique asymptote.

$$\begin{array}{r} 2x - 1 \overline{) 4x^2 - 4x + 5} \\ \underline{-(4x^2 - 2x)} \\ -2x + 5 \end{array}$$

$$\begin{array}{r} -2x + 5 \\ \underline{-(2x + 1)} \\ 4 \end{array}$$

∴ the equation of the OA is $\boxed{2x - 1}$

6. Given $h(x) = \frac{9x^2-1}{x^4}$, $h'(x) = \frac{-18x^2+4}{x^5}$ and $h''(x) = \frac{54x^2-20}{x^6}$, determine the following: (round to 2 decimal places)

a) Intercepts of the function

x -int: $0 = \frac{9x^2-1}{x^4}$ $x = \pm \sqrt{\frac{1}{9}} = \pm \frac{1}{3}$ y -int: $y = \frac{9(0)^2-1}{(0)^4} = \text{N/A}$ $\therefore x$ -int: $(-0.33, 0)$ and $(0.33, 0)$ (2)

b) All Applicable asymptotes and the behaviour around the asymptotes (be sure to use Calculus methods)

VA: $x=0$ HA: $\lim_{x \rightarrow \infty} f(x) = \frac{x^2(9 - \frac{1}{x^2})}{x^4} = \frac{9}{x^2} = 0^+$ $\lim_{x \rightarrow \infty} f(x) = 0^+$ $\lim_{x \rightarrow 0^+} f(x) = -\infty$ $\lim_{x \rightarrow 0^-} f(x) = -\infty$ No OA $\lim_{x \rightarrow -\infty} f(x) = 0^+$ (3)

c) Intervals of increase and decrease, classify and state coordinates of any local extrema

Intervals	$(-\infty, -0.47)$	$x = -0.47$	$(-0.47, 0)$	$x = 0$	$(0, 0.47)$	$x = 0.47$	$(0.47, \infty)$
$f'(x)$	$(+)$	0	$(-)$	VA	$(+)$	0	$(-)$
$f(x)$	INC	MAX	DEC	VA	INC	MAX	DEC

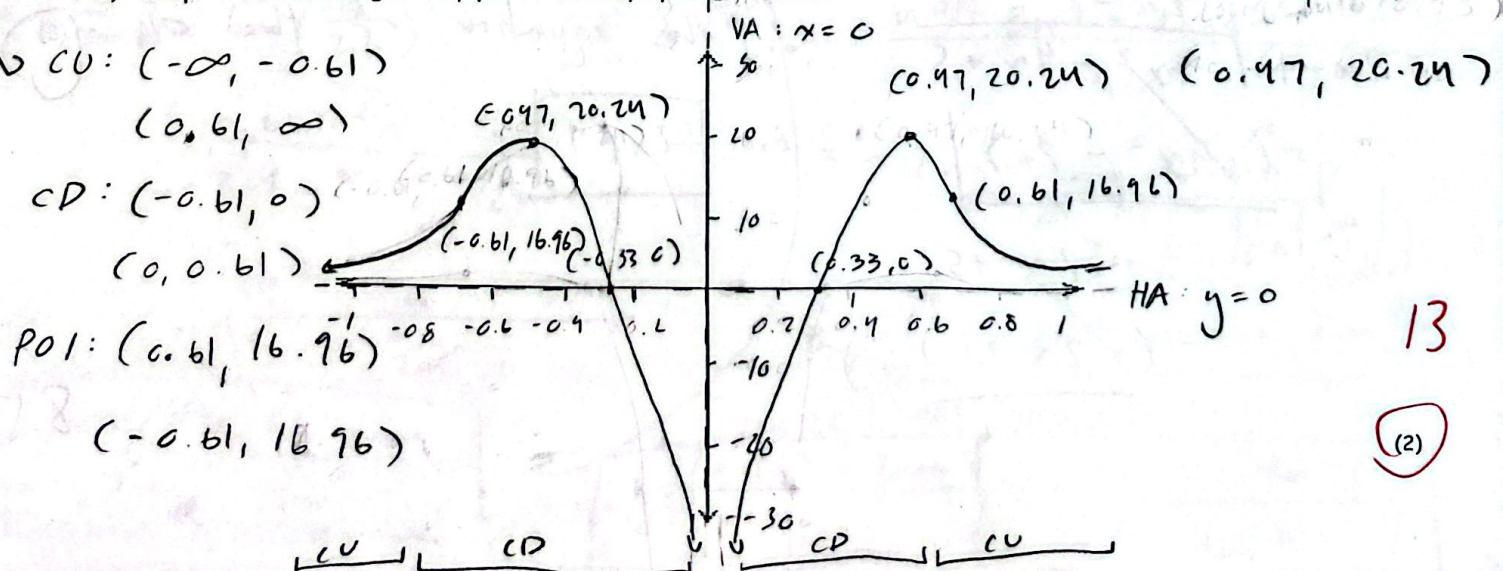
$h'(x) = \frac{-18x^2+4}{x^5}$ CP: $0 = -18x^2+4$ $x = \pm \sqrt{\frac{2}{9}} = \pm 0.47$ $x = 0, \pm 0.47$ INC: $(-\infty, -0.47)$ $(0, 0.47)$ DEC: $(-0.47, 0)$ $(0.47, \infty)$ (3)

d) Intervals of concavity and any point(s) of inflection

Intervals	$(-\infty, -0.61)$	$x = -0.61$	$(-0.61, 0)$	$x = 0$	$(0, 0.61)$	$x = 0.61$	$(0.61, \infty)$
$f''(x)$	$(+)$	0	$(-)$	VA	$(-)$	0	$(+)$
$f(x)$	CU	POI	CD	VA	CD	POI	CU

$h''(x) = \frac{54x^2-20}{x^6}$ $x = \sqrt{\frac{20}{54}} = \frac{\sqrt{30}}{3} = 0.608$ possible POI: $0, \pm 0.61$ local max: $(-0.47, 20.24)$ $(0.47, 20.24)$ (3)

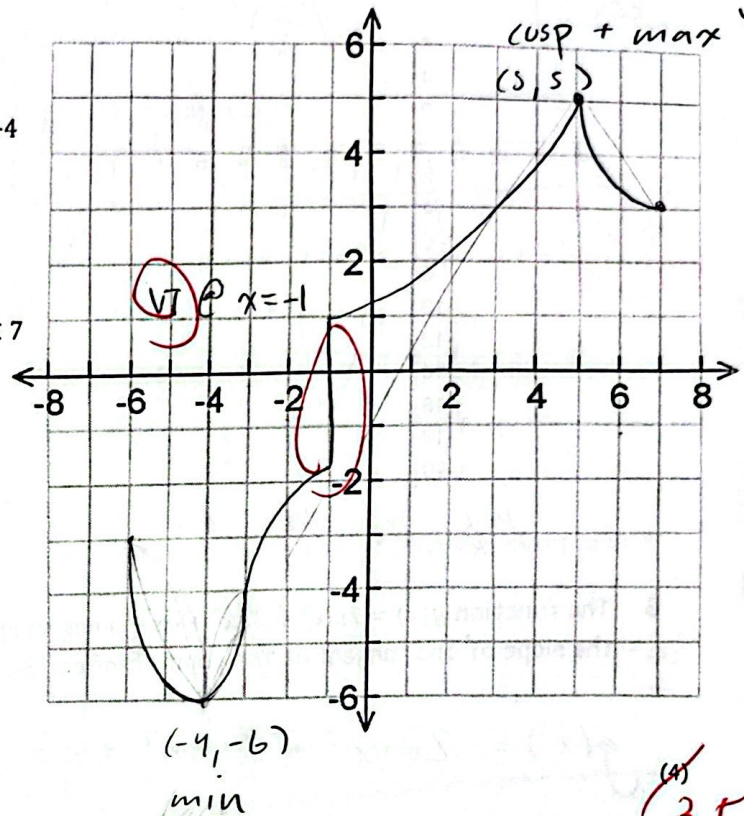
e) Graph $h(x)$ including all key points and asymptotes, labelled



Crystal be, March 28

4. Sketch a possible graph for a continuous function that has the following characteristics. Label all key features. @ $x = 5$

- $f(x)$ has a domain of $[-6, 7]$
- $f(x)$ has an absolute minimum of -6 when $x = -4$
- $f(x)$ has an absolute maximum of 5 when $x = 5$
- $f'(x) > 0$ for $-4 < x < -1$ and $-1 < x < 5$
- $f'(x) < 0$ for $-6 < x < -4$ and $5 < x < 7$
- $f''(x) < 0$ for $-3 < x < -1$ CD
- $f''(x) > 0$ for $-6 < x < -3$, $-1 < x < 5$, $5 < x < 7$
- $f'(-1) = 0$ and $f'(5) = DNE$



5. For the graph of $f'(x)$, determine the following:

a) Interval(s) of decrease for $f(x)$

$$(-\infty, -3) \cup (3, \infty)$$

b) x value of Local Maximum(s) of $f(x)$

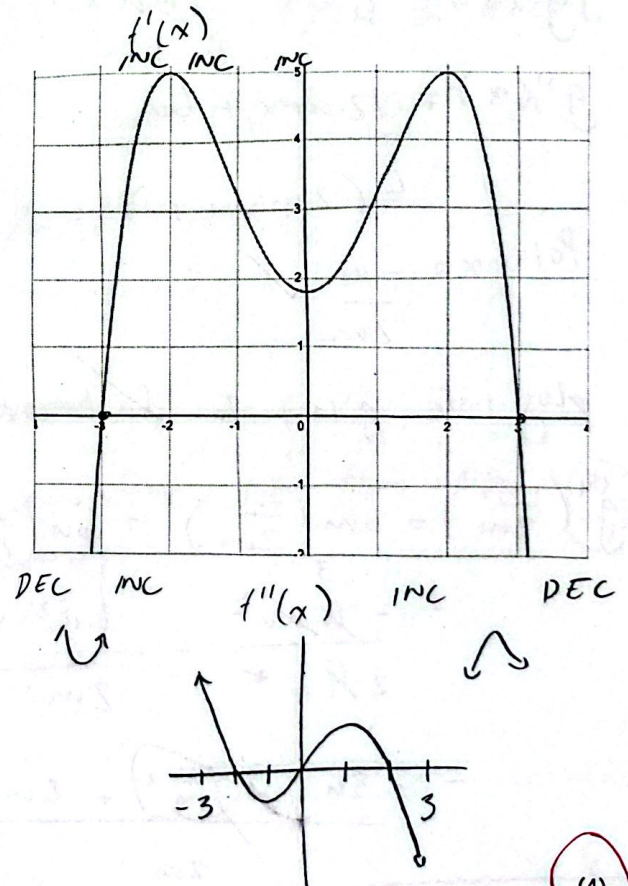
$$x = 3$$

c) Interval(s) where $f(x)$ is concave up.

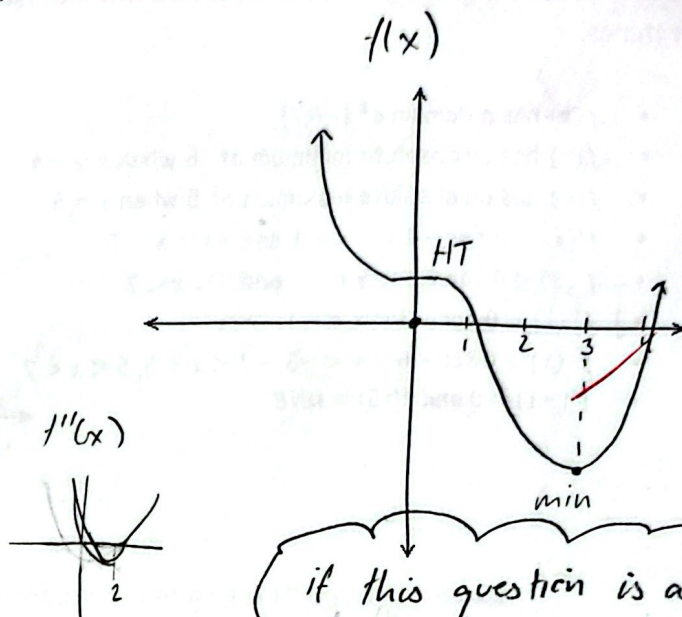
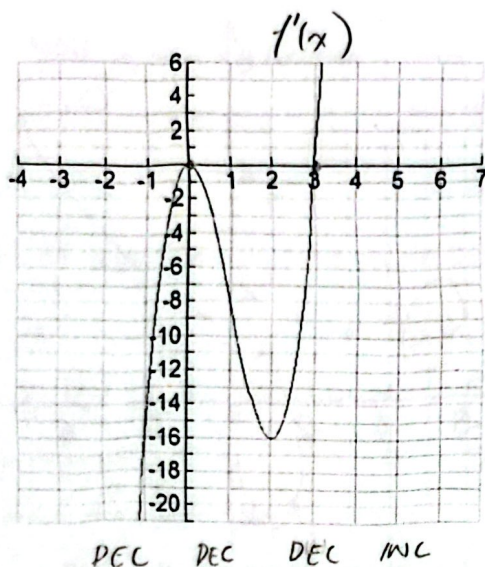
$$(-\infty, -2) \cup (0, 2)$$

d) x value of Point(s) of Inflection for $f(x)$

$$x = -2, 0, 2$$



7. Given the graph of $f'(x)$, sketch a graph of $f(x)$.



8. The function $g(x) = 2mx^3 + 3nx^2 + qx + p$ has one point of inflection. Determine a simplified expression for the slope of the tangent at the point of inflection. Assume m, n, q and p are constants.

if this question is asking only for m , then let's say $n=1$ (above + beyond)

$$g(x) = 2mx^3 + 3nx^2 + qx + p$$

$$g'(x) = 6mx^2 + 6nx + q$$

$$g''(x) = 12mx + 6n$$

$$= 6(2mx + n)$$

$$\text{Pol: } x = -\frac{n}{2m}$$

plug into $g'(x)$ to find slope

$$g'(-\frac{n}{2m}) = 6m(-\frac{n}{2m})^2 + 6n(-\frac{n}{2m}) + q$$

$$= \frac{6m \cdot n^2}{2 \cdot 4m^2} - \frac{6n^2}{2m} + \frac{q}{1}$$

$$= -\frac{3n^2}{2m} - \frac{6n^2}{2m} + \frac{2mq}{2m}$$

$$m = -\frac{9n^2 + 2mq}{2m}$$

$$g(-\frac{n}{2m}) = 2m(-\frac{n}{2m})^3 + 3n(-\frac{n}{2m})^2 + q(-\frac{n}{2m}) + p$$

$$= \frac{-2mn^3}{8m^3} - \frac{3n^3}{4m^2} - \frac{qn}{2m} + \frac{p}{1}$$

$$= -\frac{n^3}{4m^2} - \frac{3n^3}{4m^2} - \frac{2mqn}{4m^2} + \frac{4m^2p}{4m^2}$$

$$= \frac{-n^3 - 3n^3 - 2mqn + 4m^2p}{4m^2}$$

$$= \frac{-2n^3 - mqn + 2m^2p}{2m^2}$$

$$y = \left(\frac{-2n^3 - mqn + 2m^2p}{2m^2} \right) x - \frac{13n^3 + 4m^2p}{4m^2}$$

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$$y = \left(\frac{-9n^2 + 2mq}{2m} \right) x - \frac{13n^3 + 4m^2p}{4m^2}$$